

Part 1 – Project Proposal

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Degree Pathway (course)	BSc (Hons) Computer Science
Supervisor Name	Mr. Chan Sri Manukalpa
Title of Project	Personalized Tourism Recommendation and Low Commission Booking System for Community Experiences
Abstract of the project	The Sri Lankan tourism industry faces a dual challenge: high commission fees charged by international booking platforms that erode hotel profitability, and the limited digital visibility of rural cultural and sports-based tourism experiences. Existing digital platforms rarely address these interconnected local needs simultaneously. Addressing these challenges is crucial for fostering a sustainable tourism ecosystem. Reducing financial leakage strengthens local businesses, while promoting rural experiences drives economic growth and preserves cultural heritage. However, research on integrated platforms designed to tackle these problems using advanced computational techniques remains limited. The primary objective of this study is to design an AI-driven tourism application that enables Sri Lankan hotels to mitigate high commission fees while concurrently promoting cultural village events and adventure tourism nationwide. The proposed system employs a hybrid recommendation engine combining collaborative and content-based filtering, geo-aware services for location-based discovery, and an intelligent chatbot utilizing a pre-trained BERT model for natural language interaction. Local stakeholders, hoteliers, event organizers, and tour operators can list services for a nominal commission of approximately 3 percent. To ensure transparency and build user trust, the system integrates Explainable AI techniques, specifically SHAP and LIME, making the recommendation process interpretable. Data on user preferences and behavior will inform the hybrid filtering models, while the BERT-based chatbot facilitates interactive travel planning. The integration of SHAP and LIME provides post-hoc explanations for recommendations, enhancing user confidence in the system. The proposed system is expected to reduce operational costs for local providers, increase digital visibility for rural tourism offerings, and enhance user satisfaction through personalized, transparent interactions. The combination of hybrid filtering, conversational AI, and explainable methods addresses critical gaps in existing travel platforms. For practitioners, this system offers a pathway to increase direct bookings through transparent and personalized technology. For researchers, it provides a framework for applying hybrid recommender systems and explainable AI within developing tourism economies. Ultimately, this study contributes a model that leverages collaborative filtering, BERT-based conversation agents, and XAI techniques to reinforce local

	economic growth, advance cultural heritage, and deliver highly personalized travel experiences.
Project deliverables	System Components • AI-Based Dynamic Tour Recommendation Engine • Personalized Travel Planning System (Trained Machine Learning Model) • NLP-Based AI Chatbot Assistant • Sri Lanka–Oriented Location-Based Recommendation System • Web Application Interface ○ User Dashboard ○ Hotel Owner Dashboard ○ Administrator Dashboard • Mobile Application (Future Extension – Same Backend Database) • Low-Commission Reservation Management System (3%) • Centralized Tourism Database ○ Hotels ○ Village Cultural Activities ○ Sports Tourism Packages • Explainable AI (XAI) Integration ○ SHAP ○ LIME Technical & Documentation Deliverables • Fully Documented APIs • System Architecture Diagrams • GitHub Repository (Complete Source Code) • Testing and Performance Evaluation Report Academic Documentation i) Project Proposal ii) Contextual / Literature Review Report iii) Final Thesis iv) User Guide
	<i>Existing System</i> The existing world-renowned tourist services like Booking.com and Airbnb are charging large commission rates of up to 15 to 30 percent, which decreases the profits of the accommodation and activity providers. These websites mainly provide accommodation posts, and tours to rural villages, adventure sports, and community-localized cultural activities have a small presence, which are the major sources of sustainable local tourism (Gretzel et al., 2020). Moreover, the majority

Description of your artefact	of the current platforms do not have sophisticated AI-based personalization and cannot process past traveling patterns of users to offer valuable and applicable recommendations. This is a weakness that minimizes the user interaction and ignores the need to boost local economies by means of local tourism promotion (Ricci, Rokach and Shapira, 2015; Zhang et al., 2019). Aim The objective of the project is to build an AI-driven tourism services platform that will lower the commission rates to around 3 percent, offer customized traveling experiences and actively assist Sri Lankan village communities, cultural festivals, and sports tourism organizers by utilizing smart recommendation platforms. Key Objectives • To build the machine-learning-based personalized recommendation system based on the user behavior and travel history (He et al., 2017; Zhang et al., 2019). • To create a chatbot based on NLP and make it communicate with the users as a virtual travel companion (Devlin et al., 2019). • To lower commission fees charged by the platform to 3% to make tourism services more affordable and lucrative to the local providers. • To offer owner-of-hotel and activity dashboards to give real-time pricing, availability and inventory management. • To apply Explainable AI models including SHAP and LIME to elucidating and clarifying the outcomes in the recommendations to the users (Lundberg and Lee, 2017; Ribeiro, Singh and Guestrin, 2016). • To come up with a scalable back end architecture that could support both web and mobile software. List of Features • Self-learning tour planner with the use of AI. • Individual suggestions made in terms of user history and preferences. • Location-based services in Sri Lanka.
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- California Village culture and calendar of events.
- Adventure modules and sports tourism.
- Reservation and payment system that is integrated.
- Price and stock control owner dashboards.
- Admin moderation tools and statistics.

References

1. Devlin, J., Chang, M.-W., Lee, K. and Toutanova, K. (2019) BERT: Pre-training of deep bidirectional transformers language understanding, NAACL-HLT 2019, Minneapolis, USA. Association to Computational Linguistics, pp. 417186.
2. Gretzel, U., Sigala, M., Xiang, Z. and Koo, C. (2020) 'Smart tourism: foundations and developments, *Electronic Markets*, 30(1), pp. 114.
3. He, X., Liao, L., Zhang, H., Nie, L., Hu, X. and Chua, T.-S. (2017) Neural collaborative filtering, *Proceedings of the 26 th International World Wide Web Conference (WWW 2017)*, Perth, Australia. ACM, pp. 173–182.
4. Lundberg, S.M. and Lee, S.I. (2017) A unified approach to model predictions interpretation, *Advances in Neural Information Processing Systems*, 30, pp. 47654774.
5. Ricci, F., Rokach, L. and Shapira, B. (2015) *Recommender systems handbook*. 2 nd edn. Boston: Springer.
6. Ribeiro, M.T., Singh, S. and Guestrin, C. (2016) 'Why should I trust you? The predictions of any classifier explained, *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, San Francisco, USA. ACM, pp. 1135–1144.
7. Zhang, S., Yao, L., Sun, A. and Tay, Y. (2019) 'Deep learning based recommender system: A survey and new perspectives, *ACM Computing Surveys*, 52(1), pp. 138).

Risk analysis	<table><tr><th>Risk</th><th>Impact</th><th>Likelihood</th><th>Mitigation</th></tr><tr><td>AI Model Accuracy Issues</td><td>High</td><td>Moderate</td><td>Train with diverse datasets, hyperparameter tuning</td></tr><tr><td>Limited Tourism Dataset</td><td>Moderate</td><td>High</td><td>Use public datasets + synthetic data generation</td></tr><tr><td>System Scalability Issues</td><td>High</td><td>Moderate</td><td>Use cloud hosting and optimized database indexing</td></tr><tr><td>Data Privacy Concerns</td><td>High</td><td>Moderate</td><td>HTTPS encryption, secure authentication, GDPR compliance</td></tr></table>	Risk	Impact	Likelihood	Mitigation	AI Model Accuracy Issues	High	Moderate	Train with diverse datasets, hyperparameter tuning	Limited Tourism Dataset	Moderate	High	Use public datasets + synthetic data generation	System Scalability Issues	High	Moderate	Use cloud hosting and optimized database indexing	Data Privacy Concerns	High	Moderate	HTTPS encryption, secure authentication, GDPR compliance
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How does your project relate to your degree course and build upon the units/knowledge you have studied/acquired	This project is a compilation of all machine learning, artificial intelligence, database infrastructure, web development, software engineering, cloud computing, and cybersecurity concepts that we had learned in our Software Engineering degree. I am using those theories to address a practical issue of the tourism sector.																				
Resources required in developing the artefact	<table><tr><th>Category</th><th>Resources</th></tr><tr><td></td><td></td></tr></table>	Category	Resources																		
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	Programming Languages	• Python (AI & Machine Learning), JavaScript (Backend & Frontend), TypeScript (Optional), • SQL/MongoDb (Database Queries)
	Frameworks & Tools	• Node.js, Express.js (Backend Development), • TensorFlow, PyTorch, Scikit-learn (Machine Learning), • HuggingFace Transformers (BERT for NLP Chatbot), • SHAP, LIME, Grad-CAM (Explainable AI), • Pandas, NumPy (Data Processing), OpenCV (Image Processing – optional future), • React.js (Frontend), • PostgreSQL / MongoDB (Database), RESTful APIs, • JWT Authentication
	Hardware Components	• Laptop (Development Machine), • Android Mobile Device (Application Testing), • iOS Device (Optional Testing), • Internet Router / Wi-Fi Connection
	Development Tools	• Visual Studio Code, PyCharm (AI Development), • Git & GitHub (Version Control), Postman (API Testing), • Docker (Optional Containerization), Firebase / Cloud Hosting Services
Have you completed & submitted your ethics form?	YES	
If the project is a development of previous work by yourself or others, give details below. Failing to declare such previous work here may be treated as an academic offence		

Supervisor Signature:

After the proposal has been signed off by both the supervisor and course coordinator scan the proposal and upload on BREO with signatures. Projects that follow proposals that have not been approved may be cancelled and there will be no compensation for any time lost

Part 2 – List of relevant resources

Fill in this section after your project proposal has been approved by your supervisor. Use Harvard referencing (see <https://lrweb.beds.ac.uk/a-guide-to-referencing>). Modify the list below as appropriate. This list is part of Assignment 1 and will be submitted with the Project Proposal.

1. Books

- a. Aggarwal, C.C. (2016) Recommender Systems: The Textbook. Cham: Springer.
- b. Géron, A. (2019) Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow. 2nd edn. Sebastopol: O'Reilly Media.
- c. Goodfellow, I., Bengio, Y. and Courville, A. (2016) Deep Learning. Cambridge, MA: MIT Press.
- d. Ricci, F., Rokach, L. and Shapira, B. (eds.) (2015) Recommender Systems Handbook. 2nd edn. Boston: Springer.

2. Journal Papers

- a. Zhang, S., Yao, L., Sun, A. and Tay, Y. (2019) 'Deep learning based recommender system: A survey and new perspectives', ACM Computing Surveys, 52(1), pp. 1–38.
- b. Gretzel, U., Sigala, M., Xiang, Z. and Koo, C. (2020) 'Smart tourism: Foundations and developments', Electronic Markets, 25(3), pp. 179–188.
- c. He, X., Liao, L., Zhang, H., Nie, L., Hu, X. and Chua, T.S. (2017) 'Neural collaborative filtering', Proceedings of the International World Wide Web Conference, pp. 173–182.
- d. Devlin, J., Chang, M.W., Lee, K. and Toutanova, K. (2019) 'BERT: Pre-training of deep bidirectional transformers for language understanding', Proceedings of NAACL-HLT 2019, pp. 4171–4186.
- e. Ribeiro, M.T., Singh, S. and Guestrin, C. (2016) 'Why should I trust you? Explaining the predictions of any classifier', Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 1135–1144.

3. Web Sites with relevant information

- a. TensorFlow (2025) TensorFlow Documentation. Available at: <https://www.tensorflow.org> (Accessed: 24 February 2026).
- b. PyTorch (2025) PyTorch Documentation. Available at: <https://pytorch.org> (Accessed: 24 February 2026).
- c. Scikit-learn (2025) Scikit-learn User Guide. Available at: <https://scikit-learn.org> (Accessed: 24 February 2026).

- d. *Hugging Face (2025) Transformers Documentation. Available at: <https://huggingface.co/docs> (Accessed: 24 February 2026).*
- e. *OpenCV (2025) OpenCV Documentation. Available at: <https://docs.opencv.org> (Accessed: 24 February 2026).*

Relevant software

- a. *Python 3.10+ – Core programming language for AI and backend development.*
- b. *TensorFlow / Keras – Deep learning model development.*
- c. *PyTorch – Alternative deep learning framework.*
- d. *Scikit-learn – Machine learning algorithms and preprocessing. Pandas and NumPy – Data preprocessing and manipulation.*
- e. *Flask or FastAPI – Backend API development.*
- f. *PostgreSQL or Firebase – Database management system.*
- g. *Visual Studio Code / PyCharm – Development IDE.*
- h. *Git and GitHub – Version control system.*
- i. *Postman – API testing tool.*
- j. *SHAP and LIME – Explainable AI tools.*
- k. *Docker (optional) – Containerization and deployment.*

4. Relevant hardware

- a. *Development Laptop (Minimum 16GB RAM recommended for ML training).*
- b. *Cloud Server (AWS / Google Cloud / Firebase Hosting).*
- c. *Internet Router / Network Access.*
- d. *External Storage (Dataset backup).*

5. Other

- a. *Kaggle (2025) Tourism and recommendation datasets. Available at: <https://www.kaggle.com> (Accessed: 24 February 2026). Google Scholar – Academic paper search engine for AI and tourism research.*
- b. *IEEE Xplore Digital Library – Research database for AI, ML and system design papers.*
- c. *Scopus Database – Indexed journal access.*
- d. *Web of Science – Citation indexed research database.*